

Sampling What We Cannot Sum

Markov Chain Monte Carlo at the Frontier

One idea from Lecture 4, a walker that **never computes the impossible sum** Z , is behind black-hole measurements, image generators and a new way to sample language models. Today we follow it to the frontier.

Yuwen Huang

Advanced Machine Learning and Deep Learning · HKUST (Guangzhou)

Builds on Lecture 4: Sampling replaces an impossible sum

Outline

► The walker of Lecture 4

Troubles 1–3: the walker measures

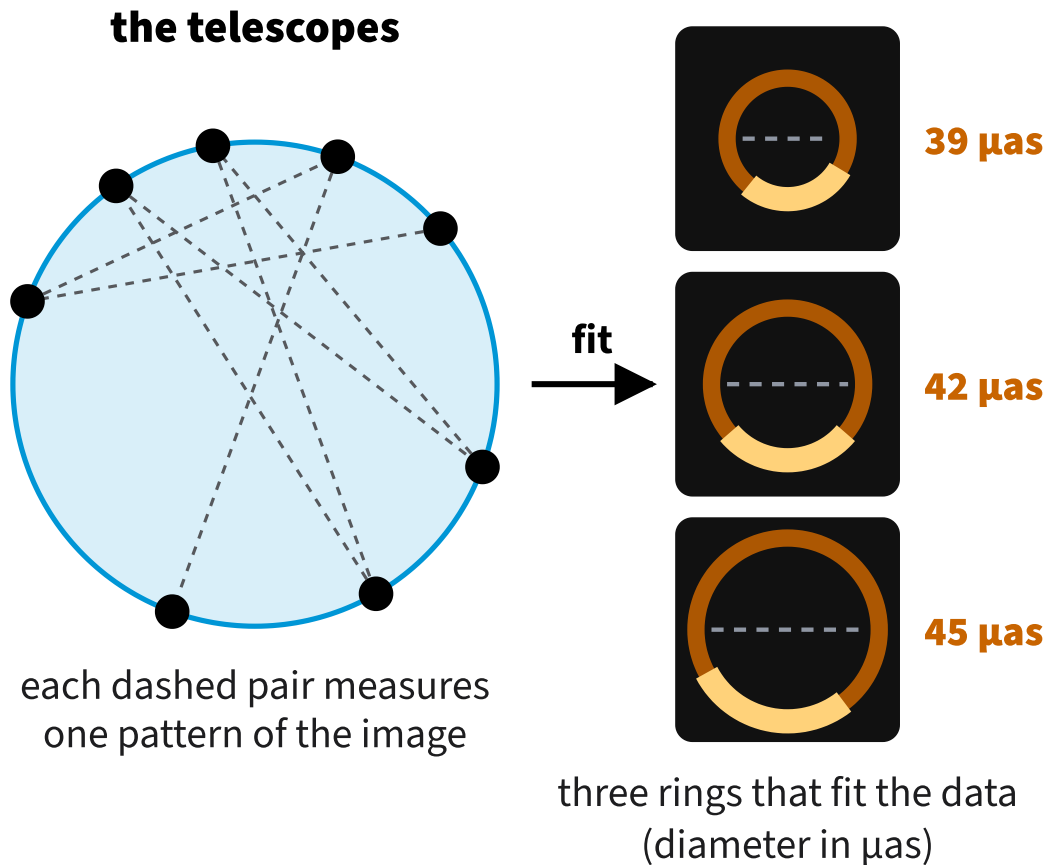
Troubles 4–5: the walker creates

Conclusion

One story: a walker that never computes Z , the troubles it meets at the frontier, and the small change that fixes each. First, a black-hole picture and the trick of Lecture 4.

The walker of Lecture 4

April 2019: the first picture of a black hole.



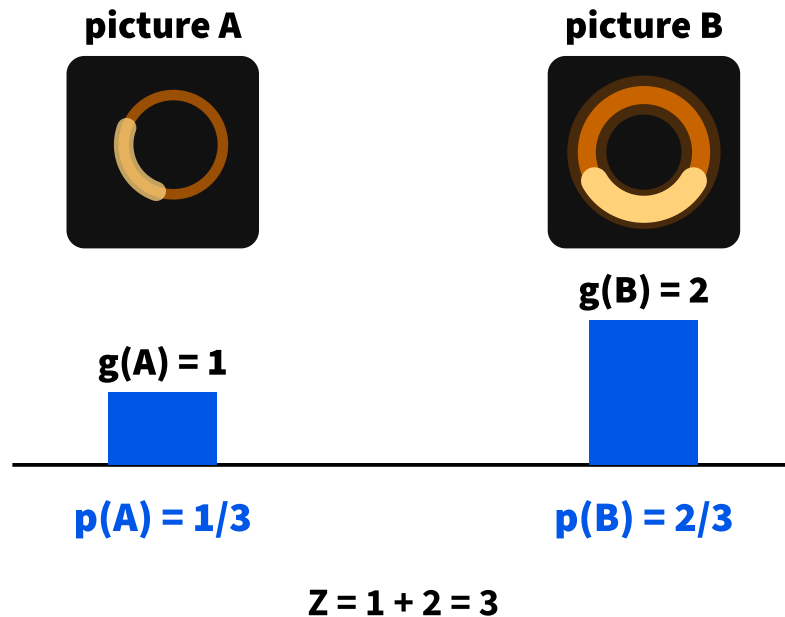
Look: telescopes across the Earth gave only a few measurements, and all three rings on the left fit them.

Q: Which picture is right?

A: Many fit. Draw them all, each as often as it fits.

The walker of Lecture 4

Main idea: draw each picture x as often as it fits.



$$p(x) = \frac{g(x)}{Z}$$

← fit: easy for one picture

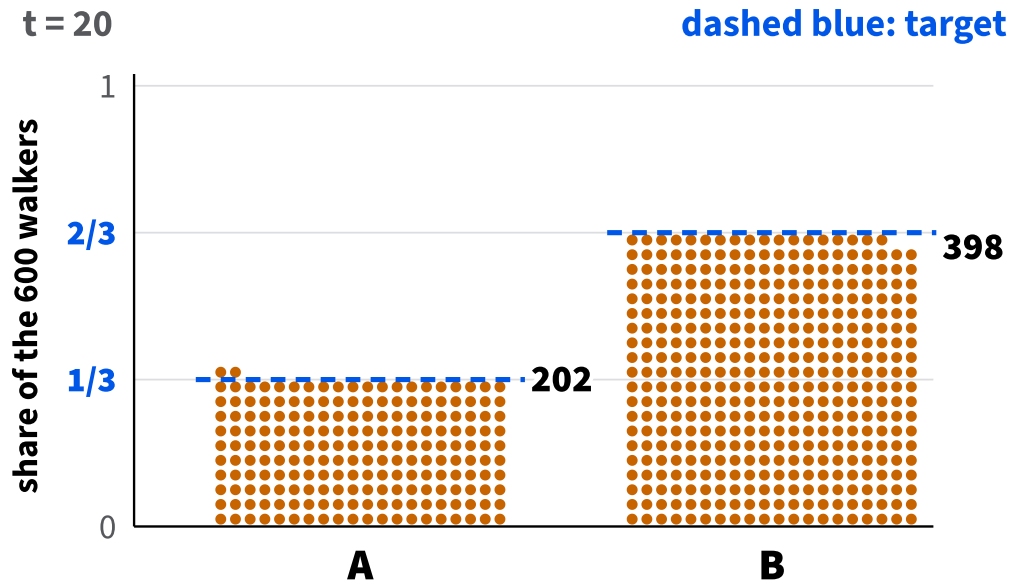
← sum over all pictures: hard

Example Fits 1 and 2: draw A and B $\frac{1}{3}$ and $\frac{2}{3}$ of the time.

Look: here $Z = 1 + 2 = 3$, but for a real image the sum has more terms than there are atoms in the universe.

The walker of Lecture 4

The trick: compare two pictures, and Z cancels.



$$\frac{p(\text{new})}{p(\text{old})} = \frac{g(\text{new})/\cancel{Z}}{g(\text{old})/\cancel{Z}} = \frac{g(\text{new})}{g(\text{old})}$$

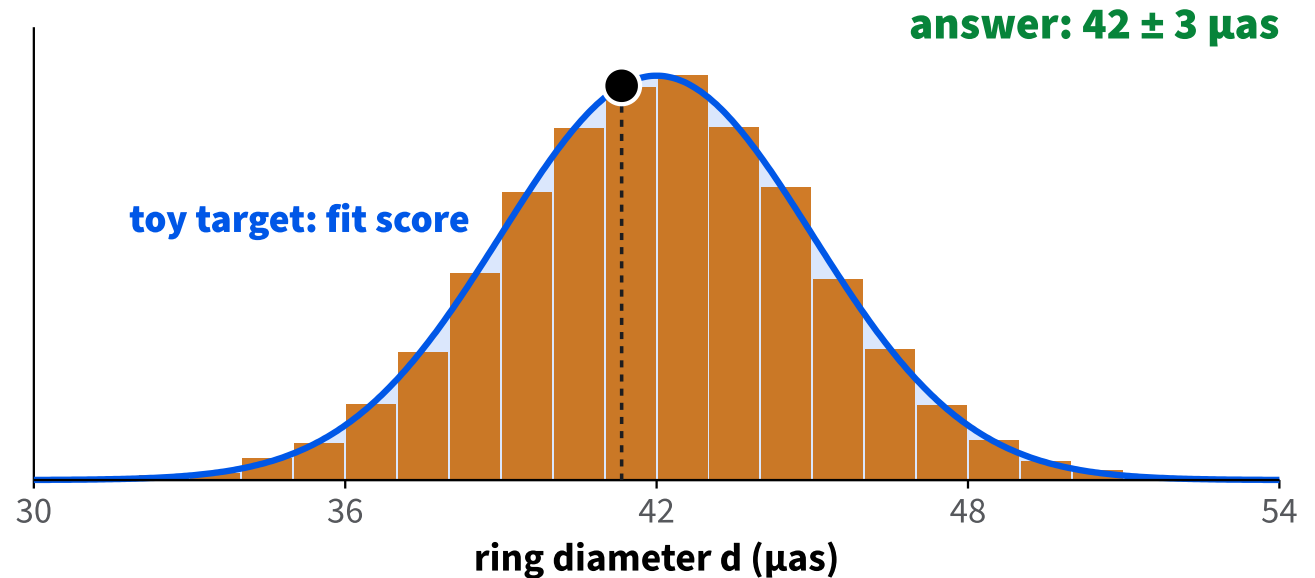
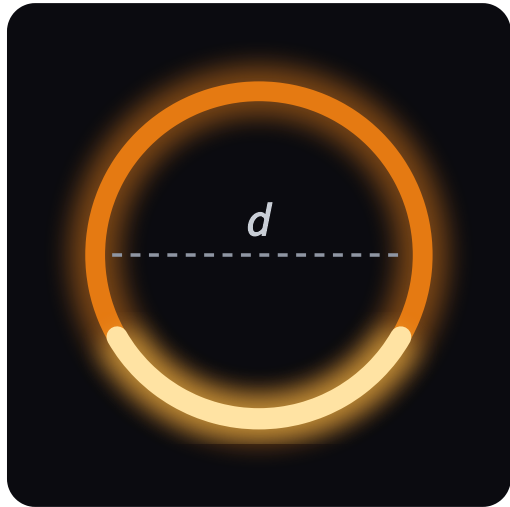
The test [Metropolis et al., 1953; Hastings, 1970]:

- ▶ **Uphill** ($A \rightarrow B$): always move.
- ▶ **Downhill** ($B \rightarrow A$): move with probability $g(\text{new})/g(\text{old}) = \frac{1}{2}$.

Watch: 600 walkers start at A and keep hopping; the shares settle close to $\frac{1}{3}$ at A and $\frac{2}{3}$ at B, and nobody computed Z .

The walker of Lecture 4

The walker measured the black-hole ring.



Watch: with the same test (p. 5), the walker steps between nearby diameters; the ring follows it, and its visits pile up under the target.

Result: diameter 42 ± 3 microarcseconds (μas), so the black hole weighs 6.5 ± 0.7 billion Suns [Event Horizon Telescope Collaboration, 2019].

Main Question

The walker never computes Z .

Can it also weigh colliding black holes, track COVID-19 lockdowns, give neural networks error bars, paint pictures, and help language models reason?

This is indeed the case, after a few small changes.

Keep this question in mind: each part changes the same walker in one small way.

The walker of Lecture 4

Five troubles at the frontier, five fixes.

	Trouble	Fix	Frontier
1	far-apart hills	???	colliding black holes
2	many dimensions	???	COVID-19 lockdowns
3	too much data	???	neural-network error bars
4	no formula, only examples	???	image generators
5	a sum over all texts	???	language-model reasoning

Read: troubles 1–3 appear when the walker measures, 4–5 when it creates. The fixes are revealed one by one.

Outline

The walker of Lecture 4

► **Troubles 1–3: the walker measures**

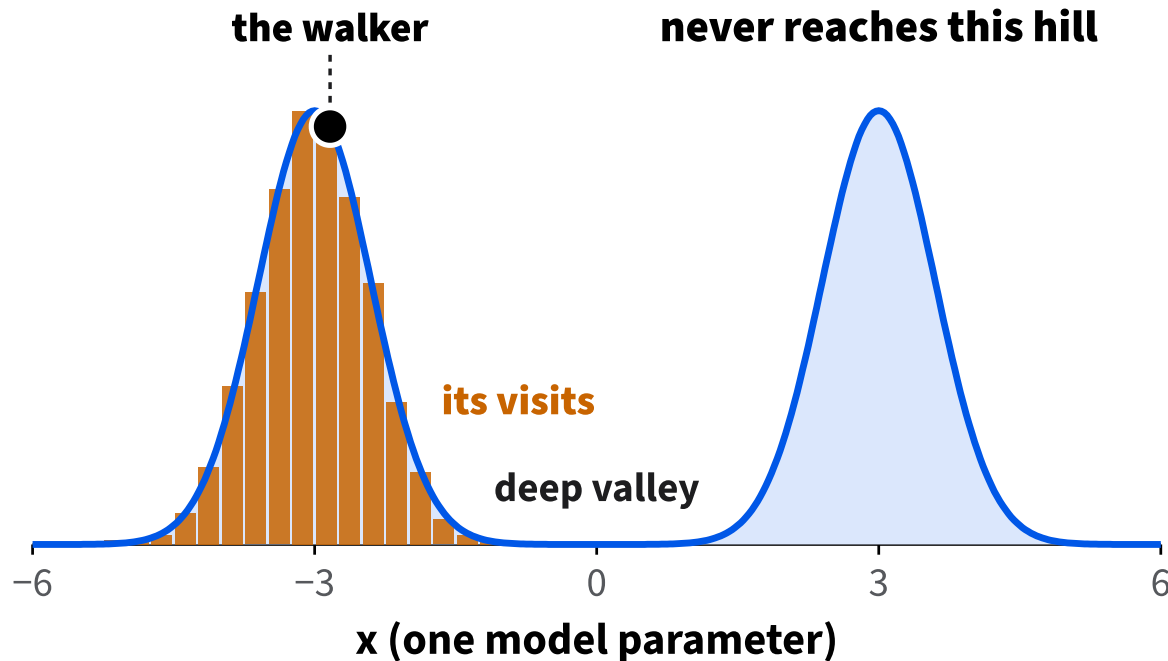
Troubles 4–5: the walker creates

Conclusion

Next: three ways the walker fails when it measures, and the small change that fixes each.

Trouble 1: far-apart hills

Trouble: two far-apart hills trap the walker.



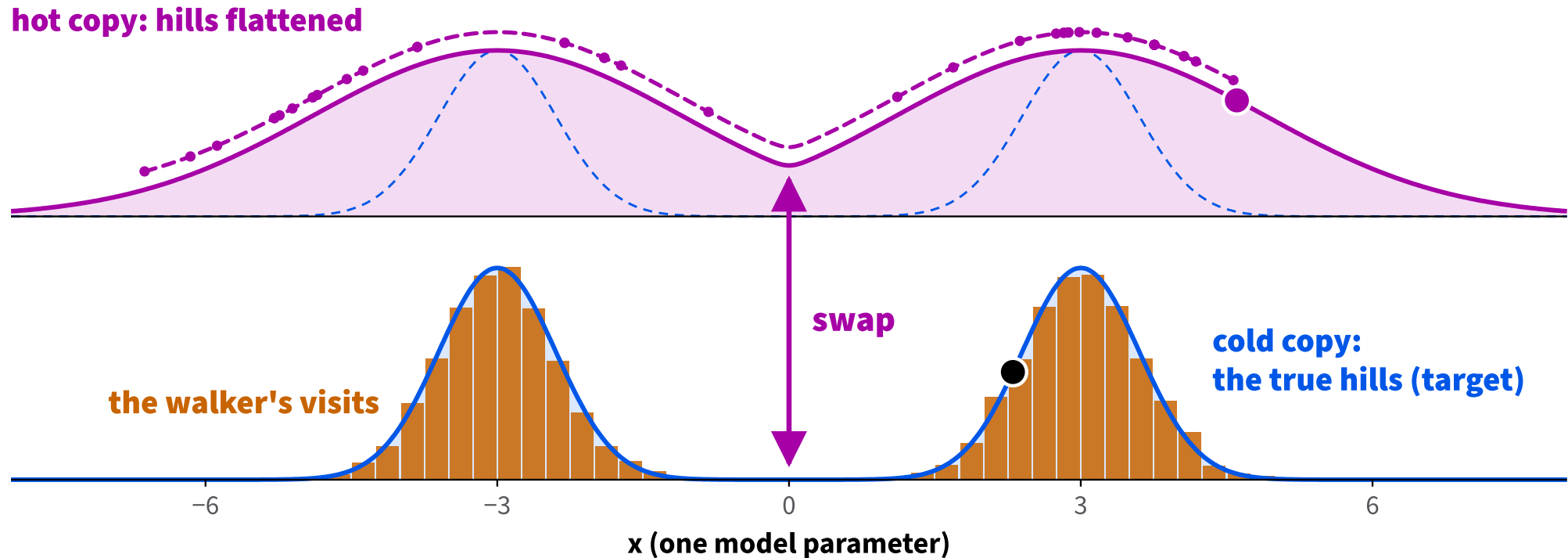
Watch: its visits fill only the left hill. The right hill, an equally good explanation, is never found, and nothing warns us.

Q: Why is it stuck?

A: Crossing needs many downhill moves in a row.

Trouble 1: far-apart hills

Fix: heat a copy until the valley is shallow; let it swap with the walker.



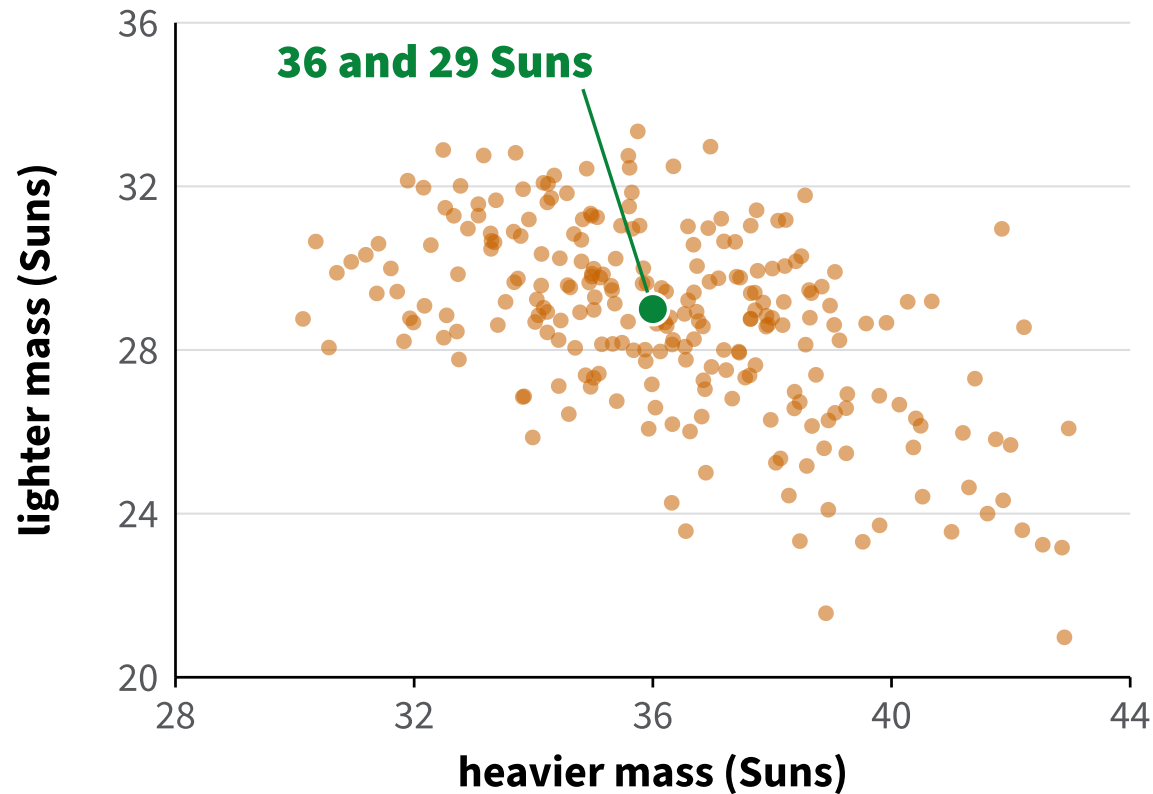
Watch: the hot copy (top, dashed: its last steps) crosses the shallow valley; at each magenta swap the copies trade places, so the walker's visits fill both hills.

This is **parallel tempering** [Swendsen and Wang, 1986; Geyer, 1991].

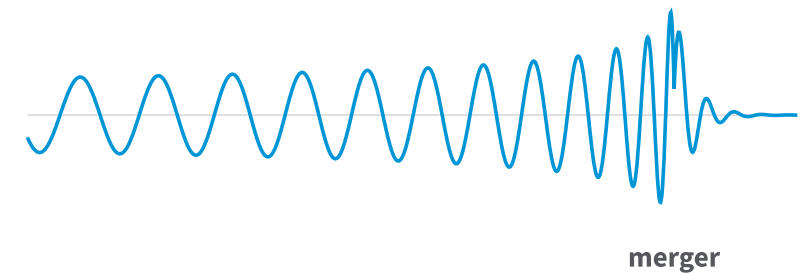
Trouble 1: far-apart hills

Result: two colliding black holes were weighed with hot and cold walkers.

each dot: one (illustrative) sample



the measured wave (schematic)

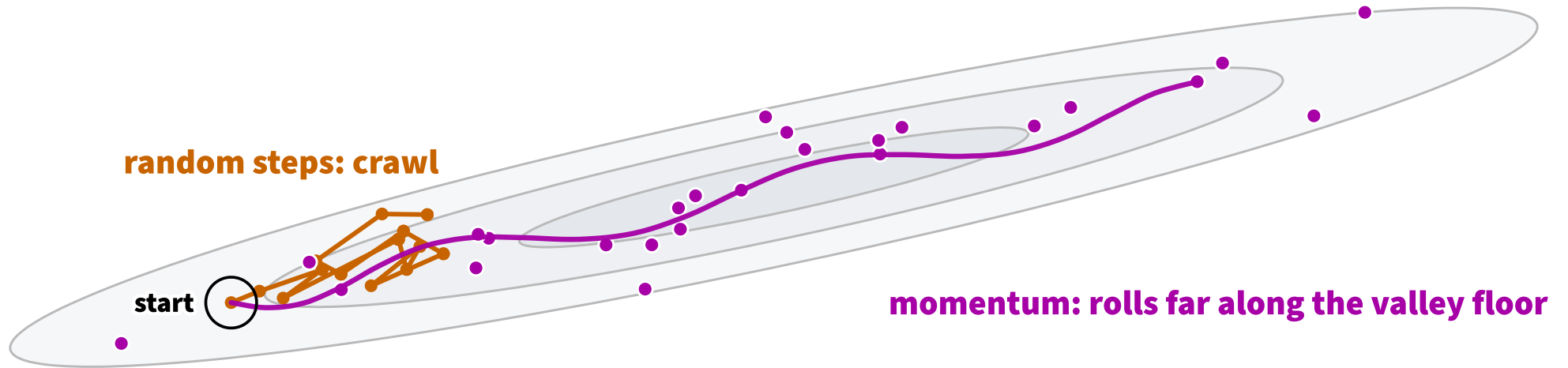


14 September 2015: a 0.2-second chirp from 1.3 billion light-years away [Veitch et al., 2015; Abbott et al., 2016].

Look: each dot is one sample of the two masses; the cloud's centre is the estimate, its size the error bar.

Trouble 2: many dimensions

Trouble: random steps crawl. **Fix:** turn the hill into a valley and roll a ball along it.



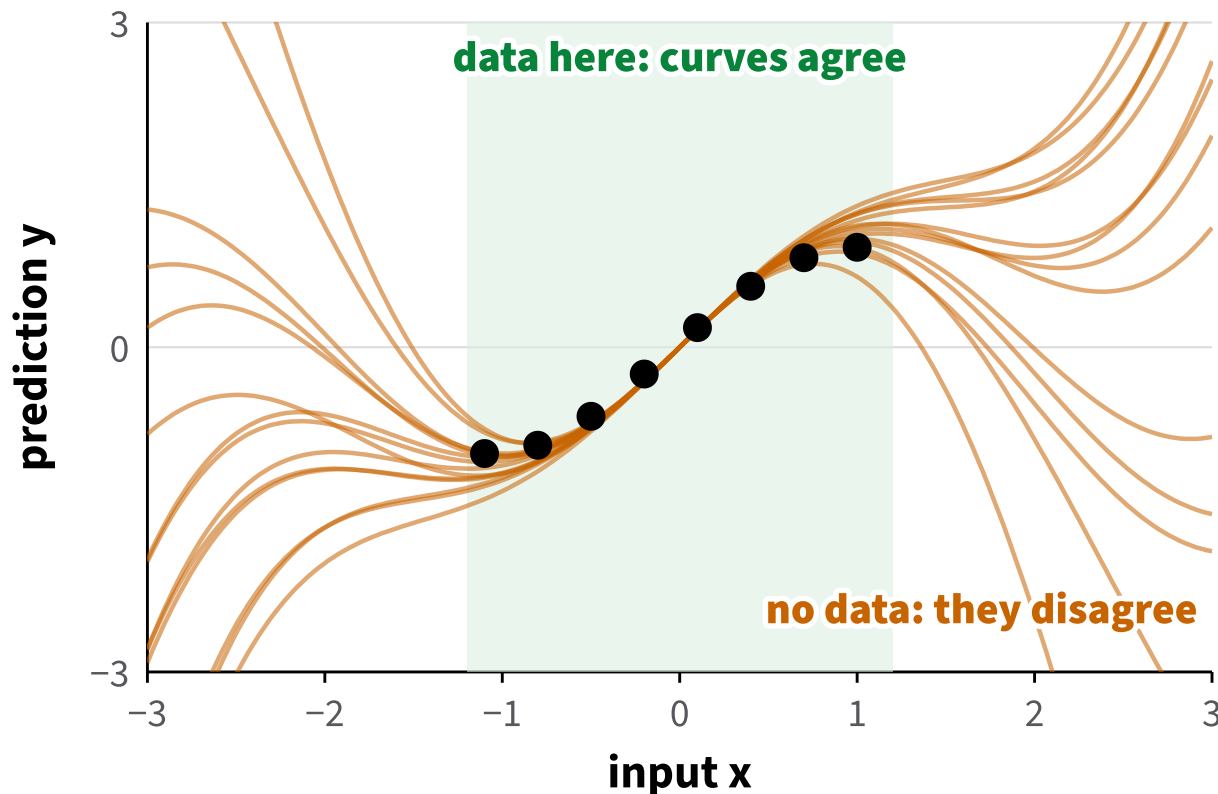
Watch: in 25 moves each, the orange random walker barely leaves the start, while the magenta ball, keeping its energy, swings along the whole valley.

This is **Hamiltonian Monte Carlo** [Duane et al., 1987]; Stan used it for COVID-19 lockdowns [Flaxman et al., 2020].

Trouble 3: too much data

Walk over a network's weights, and it knows where it is unsure.

each orange curve: one sampled network (schematic)

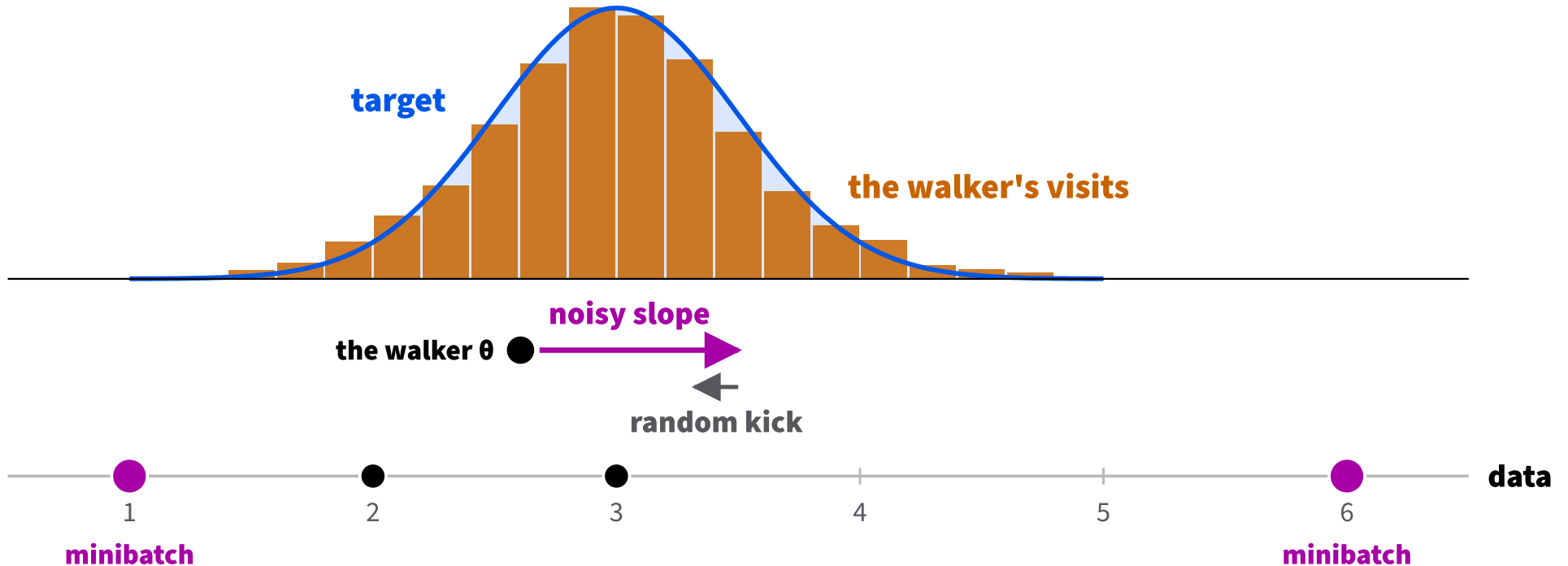


Look: near the data the sampled networks agree; far from it they fan out. That spread says “I am not sure here”.

Trouble: each comparison scores the network on every training example, millions of them.

Trouble 3: too much data

Fix: follow a minibatch slope, add a little noise, and skip the test.



Watch: each step reads only two data points (magenta) and adds a random kick; the visits settle close to the target, slightly too wide without the test.

This is **stochastic-gradient Langevin dynamics** [Welling and Teh, 2011]; a slope, like a ratio, never needs Z .

Outline

The walker of Lecture 4

Troubles 1–3: the walker measures

► **Troubles 4–5: the walker creates**

Conclusion

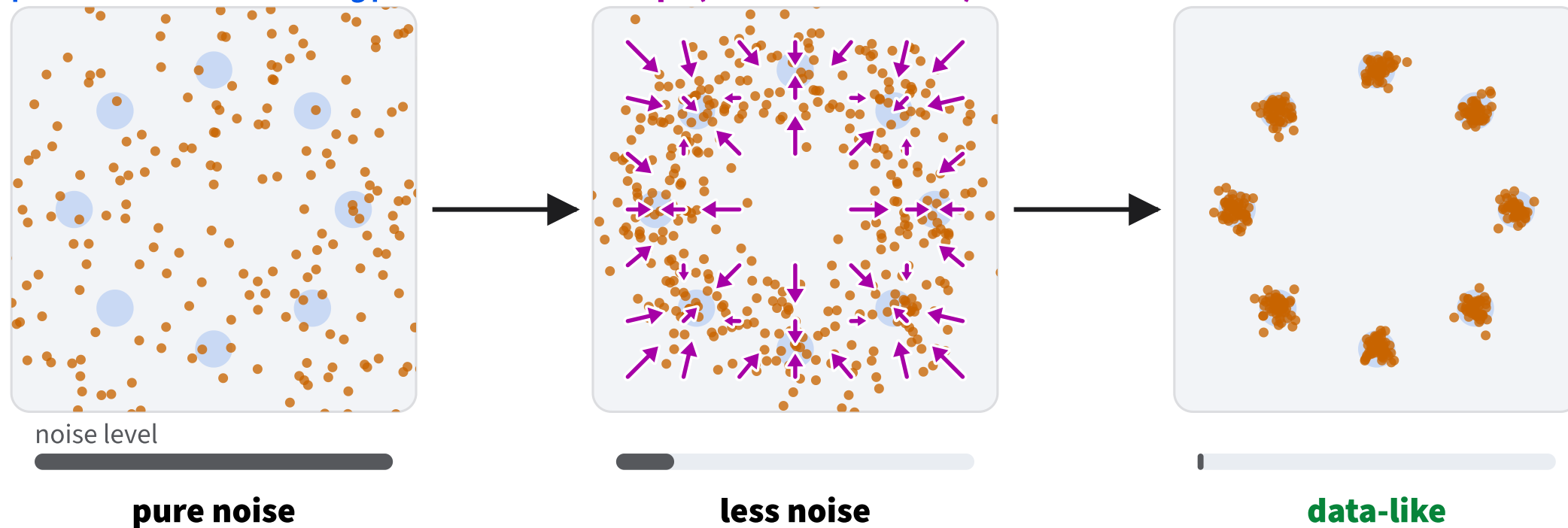
Next: two more troubles, once the walker starts creating: pictures, then text.

Trouble 4: no formula

Trouble: photos have no formula, only examples. **Fix:** learn the slope, walk from noise.

pale blue: where the training photos lie

slope (a network learns it)

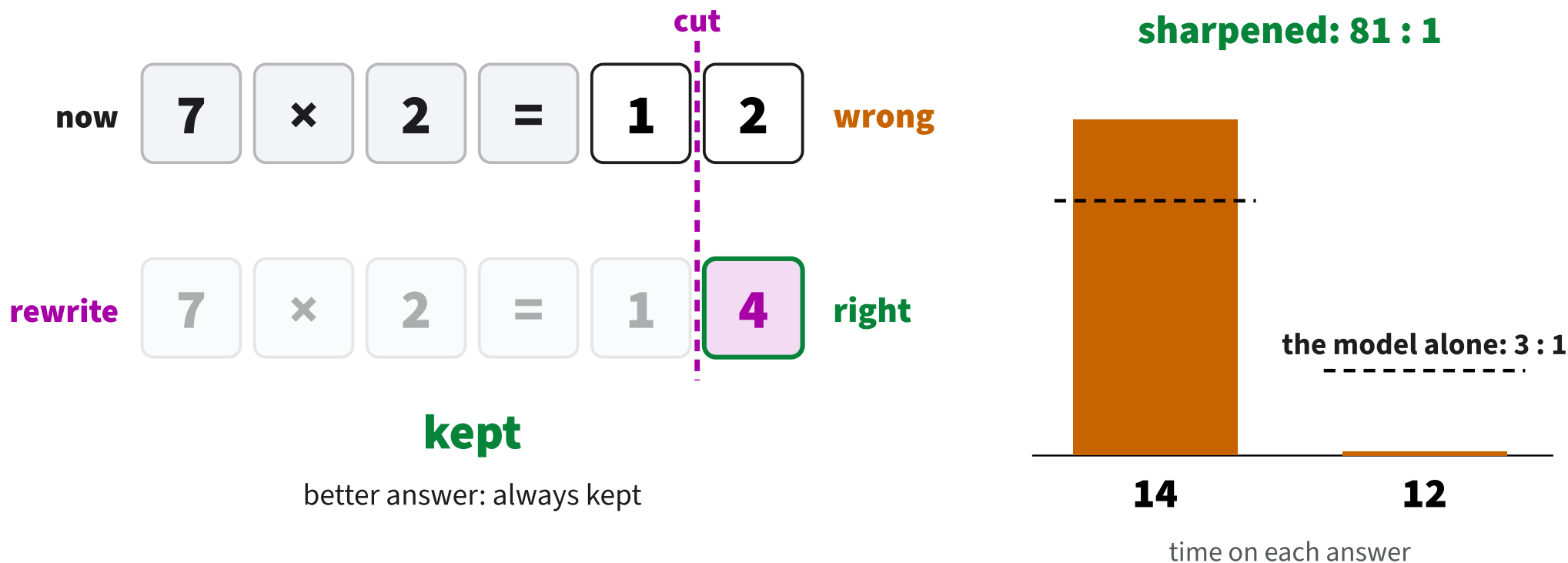


Watch: each dot is one whole picture. Following the slope (p. 15) as the noise shrinks, random dots gather where the photos lie.

This is the idea behind **diffusion image generators** [Song and Ermon, 2019; Ho et al., 2020].

Trouble 5: a sum over all texts

Trouble: sharpening sums over all texts. **Fix:** rewrite the end, compare.



Watch: sharpening makes likely answers likelier. The model rewrites the end, the test (p. 5) keeps or rejects it, and the right answer, 14, ends near 81 : 1, far above the model's own 3 : 1.

This is **power sampling**: better reasoning with no extra training, at about 9 times the cost [Karan and Du, 2025].

Outline

The walker of Lecture 4

Troubles 1–3: the walker measures

Troubles 4–5: the walker creates

► **Conclusion**

Finally: the five fixes side by side, and what is still open.

Conclusion

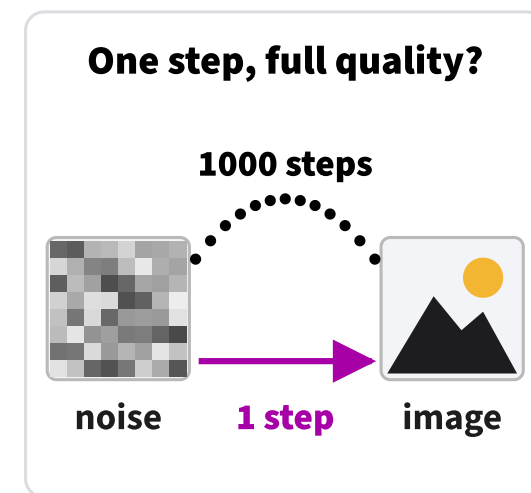
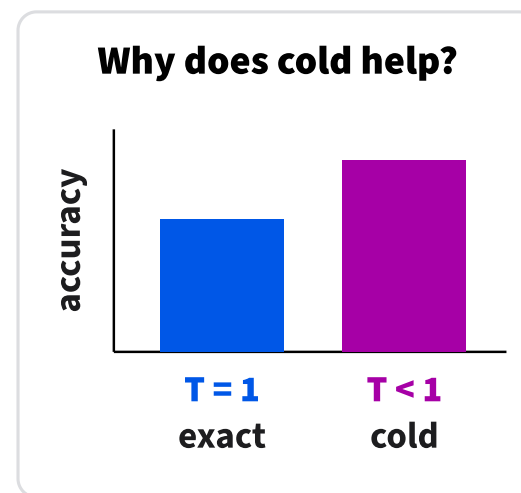
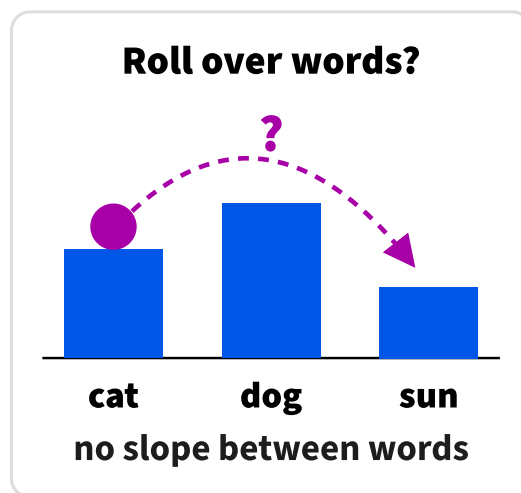
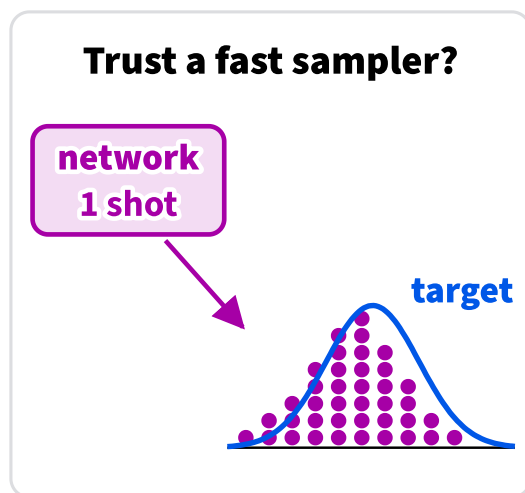
One walker, five small changes, and Z is never computed.

	Trouble	Fix	Frontier
1	far-apart hills	heat a copy, swap	colliding black holes
2	many dimensions	roll a ball (momentum)	COVID-19 lockdowns
3	too much data	minibatch slope + noise	neural-network error bars
4	no formula, only examples	learn the slope from examples	image generators
5	a sum over all texts	rewrite the end, compare	language-model reasoning

Read: fix 1 changes the copies; fixes 2–5 change the move, and fixes 3–4 also drop the test. None computes Z : a ratio of fits cancels it, and a slope never sees it.

Conclusion

- ▶ It is possible to sample **without computing Z** : compare two states, or follow the slope.
- ▶ Each frontier changes **one part** of the walker: its copies, its move, or its test.
- ▶ We leave four **open problems**:



Think: each is active research and would make a good course project; start from Dax et al. (2021), Wenzel et al. (2020) and Song et al. (2023).

Selected References I

Start here: Hastings (1970) explains the walker itself; the others are the methods and today's results.

-  [Abbott, B. P., et al. \(LIGO Scientific Collaboration and Virgo Collaboration\) \(2016\).](#)
Properties of the binary black hole merger GW150914.
Physical Review Letters, 116(24):241102.
-  [Dax, M., Green, S. R., Gair, J., Macke, J. H., Buonanno, A., and Schölkopf, B. \(2021\).](#)
Real-time gravitational wave science with neural posterior estimation.
Physical Review Letters, 127(24):241103.
-  [Duane, S., Kennedy, A. D., Pendleton, B. J., and Roweth, D. \(1987\).](#)
Hybrid Monte Carlo.
Physics Letters B, 195(2):216–222.
-  [Event Horizon Telescope Collaboration \(2019\).](#)
First M87 Event Horizon Telescope results. VI. The shadow and mass of the central black hole.
The Astrophysical Journal Letters, 875(1):L6.
-  [Flaxman, S., Mishra, S., Gandy, A., et al. \(2020\).](#)
Estimating the effects of non-pharmaceutical interventions on COVID-19 in Europe.
Nature, 584:257–261.
-  [Geyer, C. J. \(1991\).](#)
Markov chain Monte Carlo maximum likelihood.
In Computing Science and Statistics: Proceedings of the 23rd Symposium on the Interface, pages 156–163.

Selected References II

Good entry points: Song and Ermon (2019) for image generation; Karan and Du (2025) for language models.



Hastings, W. K. (1970).

Monte Carlo sampling methods using Markov chains and their applications.

Biometrika, 57(1):97–109.



Ho, J., Jain, A., and Abbeel, P. (2020).

Denoising diffusion probabilistic models.

In Advances in Neural Information Processing Systems 33, pages 6840–6851.



Karan, A. and Du, Y. (2025).

Reasoning with sampling: your base model is smarter than you think.

arXiv:2510.14901.



Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., and Teller, E. (1953).

Equation of state calculations by fast computing machines.

The Journal of Chemical Physics, 21(6):1087–1092.



Song, Y. and Ermon, S. (2019).

Generative modeling by estimating gradients of the data distribution.

In Advances in Neural Information Processing Systems 32.

Selected References III

A short read: Welling and Teh (2011), eight pages on how sampling meets minibatch training.



Song, Y., Dhariwal, P., Chen, M., and Sutskever, I. (2023).

Consistency models.

In Proceedings of the 40th International Conference on Machine Learning, PMLR 202:32211–32252.



Swendsen, R. H. and Wang, J.-S. (1986).

Replica Monte Carlo simulation of spin-glasses.

Physical Review Letters, 57(21):2607–2609.



Veitch, J., Raymond, V., Farr, B., et al. (2015).

Parameter estimation for compact binaries with ground-based gravitational-wave observations using the LALInference software library.

Physical Review D, 91(4):042003.



Welling, M. and Teh, Y. W. (2011).

Bayesian learning via stochastic gradient Langevin dynamics.

In Proceedings of the 28th International Conference on Machine Learning, pages 681–688.



Wenzel, F., Roth, K., Veeling, B. S., Świątkowski, J., Tran, L., Mandt, S., Snoek, J., Salimans, T., Jenatton, R., and Nowozin, S. (2020).

How good is the Bayes posterior in deep neural networks really?

In Proceedings of the 37th International Conference on Machine Learning, PMLR 119:10248–10259.

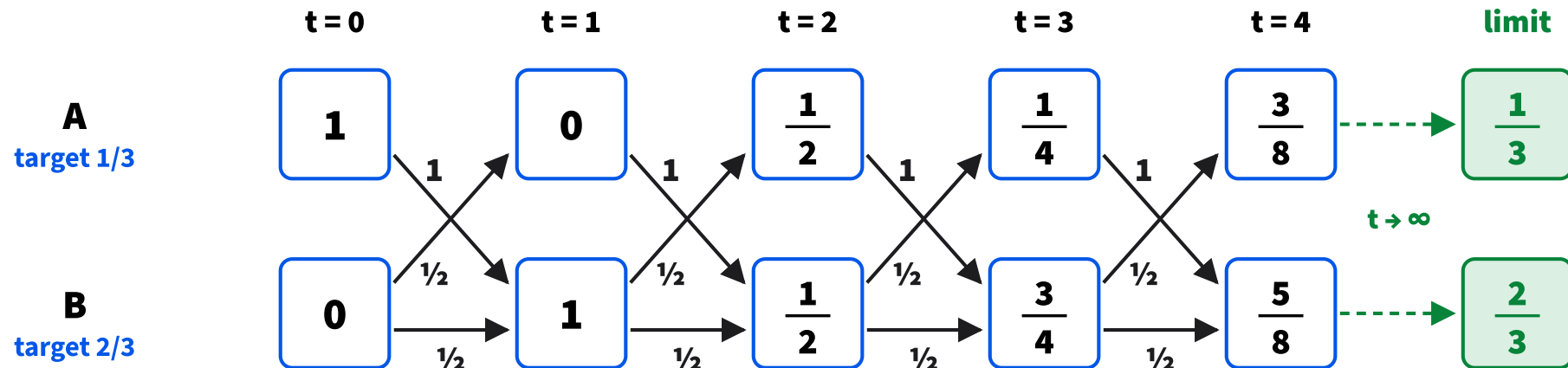
Thank you!

Next: Lab 5 (Saturday 10 October), the expectation–maximization algorithm at the frontier

Try the walker yourself: the ten-line Python version is on the backup slides that follow.

Backup: the walker of Lecture 4

Why the walkers settle at one third and two thirds: the flows balance.



Look: from A a walker always moves to B, from B back half the time. Each box adds box $\times$ arrow over the arrows into it, and the values close in on the point where the flows balance, $\frac{1}{3} \cdot 1 = \frac{2}{3} \cdot \frac{1}{2}$ (Lecture 4, Theorem 9).

Backup: the walker of Lecture 4

The whole walker is a few lines of Python, and it never uses Z .

```
import random
random.seed(0)
g = {"A": 1, "B": 2}                                # fit scores; Z is never computed
x, visits = "A", {"A": 0, "B": 0}
for t in range(100_000):
    y = "B" if x == "A" else "A"                    # propose the other picture
    if random.random() < min(1, g[y] / g[x]):        # the test: uphill always, downhill g(new)/g(old)
        x = y
    visits[x] += 1
print({s: v / 100_000 for s, v in visits.items()})
{'A': 0.33392, 'B': 0.66608}
```

Try: change the fits 1 and 2 to any two numbers; the shares follow their ratio, and Z is still never computed.

Backup: Trouble 1

The swap of parallel tempering is again a comparison.

A copy at temperature T walks on $g(x)^{1/T}$. Two copies at $T_1 < T_2$ propose to swap their states x_1, x_2 ; the test compares the swapped pair with the current pair:

$$\frac{g(x_2)^{1/T_1} \cdot g(x_1)^{1/T_2}}{g(x_1)^{1/T_1} \cdot g(x_2)^{1/T_2}} = \left(\frac{g(x_2)}{g(x_1)} \right)^{\frac{1}{T_1} - \frac{1}{T_2}}.$$

Accept with $\min\{1, \text{this ratio}\}$. In the animation $T_1 = 1$ and $T_2 = 10$, so the power is $\frac{9}{10}$.

Notice: only the ratio $g(x_2)/g(x_1)$ appears. The swap, like every step of the walker, never needs Z .

Backup: Trouble 2

One leapfrog step, by hand (a roll chains many of them).

Height $U(x) = -\ln g(x) = \frac{1}{2} \cdot x^2$, slope $U'(x) = x$. Start at $x = 1$ with momentum $v = 1$, step 0.5; energy $H = U(x) + \frac{1}{2} \cdot v^2 = 1$.

$$v = 1 - 0.25 \cdot 1 = 0.75$$

$$x = 1 + 0.5 \cdot 0.75 = 1.375$$

$$v = 0.75 - 0.25 \cdot 1.375 = 0.406$$

$$H' = \frac{1}{2} \cdot 1.375^2 + \frac{1}{2} \cdot 0.406^2 = 1.028$$

$$\min\{1, e^{-(1.028-1)}\} = 0.97$$

half kick

drift

half kick

energy almost kept

the usual test

Notice: in this one step the energy moves only from 1 to 1.028, so the test accepts it 97% of the time.